# The Road to

# Gold

## Mathematics for Competitions

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# Preface

This book is intended to help students prepare for mathematical competitions, from middle- and high-school olympiads to undergraduate competitions. It brings together mathematical theory and problem-solving methods that are especially useful for approaching challenging competition problems.

The book is organized into two parts. *Mathematical Concepts* develops relevant definitions, theorems, proofs, and applications, including advanced theory where it is useful. *Problem-Solving Strategies* focuses on methods of proof, general approaches, and practical techniques for solving difficult problems. The emphasis is on understanding not only what is true, but also why it is true and how it can be used.

This book is a companion to *Mathematics for Science Quiz*, which places greater emphasis on important results, memorable formulas, mathematical history, and interesting facts and connections. Readers looking for a broader collection of such knowledge may consult that volume; the present book provides a more detailed treatment of proofs, deeper theory, and mathematical problem solving. The two books are intended to complement, rather than replace, one another.

I hope this book helps readers develop mathematical insight, creativity, and perseverance, and encourages them to aim for the highest achievements in mathematical competitions.

# Acknowledgements

# Contents

**Preface** **1**

**Acknowledgements** **2**

**I Mathematical Concepts** **4**

**1 Algebra** **5**

**2 Number Theory** **6**

**3 Geometry** **7**

**4 Combinatorics and Probability** **8**

**5 Calculus and Analysis** **9**

**6 Linear Algebra** **10**

**II Problem-Solving Strategies** **11**

# Part I

# Mathematical Concepts

# Chapter 1

# Algebra

# Chapter 2

# Number Theory

# Chapter 3

# Geometry

# Chapter 4

# Combinatorics and Probability

# Chapter 5

# Calculus and Analysis

# Chapter 6

# Linear Algebra

# Part II

# Problem-Solving Strategies